John Kyle Cooper

Waco, Texas | Website | LinkedIn | GitHub

Professional Summary

Researcher and builder of experimental and interactive systems, with experience in eye tracking, EEG, Unity, VR, and research software. I combine experimental design, programming, and data analysis to study human perception and creative problem solving.

Education

PhD **KU Leuven**, Biomedical Sciences - Leuven, Belgium — Oct 2021 – Oct 2025

MSc **University of Rochester**, Biomedical Engineering - Rochester, New York — Aug 2019 – May 2021

BSc **Texas A&M University**, Biomedical Engineering - College Station, Texas — Aug 2015 – May 2019
- Germany Biosciences Program, Bonn, Germany
- Greece Engineering Ethics, Thessaloniki, Greece

Research and Technical Experience

IRCN, The University of Tokyo — Nov 2025 – Sept 2026
Project Researcher (Postdoctoral) · Tokyo, Japan
- Designed a creative problem-solving puzzle game and conducted experiments using eye tracking (Tobii) to help trace participants' thought processes as they worked toward solutions.
- Explored AI-supported methods for evaluating creative ideas and informing research design.

KU Leuven — Oct 2021 – Oct 2025
FWO Strategic Basic Research PhD Fellow · Leuven, Belgium
- Redesigned a Unity-based audiovisual experiment to study speech perception and human-computer interaction.
- Built neuroimaging data workflows using Python, MATLAB, R, and scikit-learn to model speech understanding.

University of Rochester — Aug 2019 – May 2021
Research Assistant · Rochester, New York
- Built a Python-based audiovisual neuroimaging setup with sub-millisecond video playback timing.
- Coordinated experimental procedures, participant scheduling, and lab operations for speech perception studies.

Texas A&M University — Aug 2016 – May 2019
Research Assistant · College Station, Texas
- Developed an Unreal Engine VR viewer for 3D cochlear scans.
- Designed and fabricated power supply boxes for optical coherence tomography experiments.

Lynntech Inc. — Aug 2018 – May 2019
Senior Design Team Member · College Station, Texas
- Designed and developed an orthotic rehabilitation device prototype.

Enmodes GmbH — Jan 2017 – May 2017
Medical Device Design Collaborator · Aachen, Germany
- Modeled a respiratory support device prototype, using Blender for design and visualization.

Research and Technical Strengths

Research Practice: Problem framing, experimental design, human-subject research, scientific writing

Programming and Analysis: Python, MATLAB, R, and JavaScript for research data analysis, automation, scripting, and experimental workflows

Experimental Systems: Audiovisual tasks, stimulus workflows, EEG systems, eye tracking (Tobii), Unity and VR environments

AI-Assisted Development: Codex and OpenCode for prototyping, coding support, requirements documentation, and context management

Collaboration: Cross-disciplinary teams, participant studies, technical documentation, presentations

Selected Publications

Objectively measuring audiovisual effects in noise using virtual human speakers Apr 2025
J. K. Cooper, J. Vanthornhout, A. van Wieringen, T. Francart
https://doi.org/10.1177/23312165251333528 (Trends in Hearing)

Neural Tracking of Audiovisual Effects in Noise Using Deep Neural Network-Generated Virtual Humans June 2025
J. K. Cooper, J. Vanthornhout, A. van Wieringen, T. Francart
https://doi.org/10.1101/2025.06.02.656280 (bioRxiv)

Large-scale multi-site study shows no association between musical training and early auditory neural sound encoding Aug 2025
K. L. Whiteford, ..., J. K. Cooper, ..., A. J. Oxenham
https://doi.org/10.1038/s41467-025-62155-5 (Nature Communications)

Communication and Outreach

IRCN, The University of Tokyo 2026
Student Outreach Instructor · Tokyo, Japan
- Led an outreach session on creativity, LLM-supported idea evaluation, and EEG as a method for studying brain activity.

KU Leuven 2024
R Programming Teaching Assistant · Leuven, Belgium
- Taught introductory R programming and statistics concepts to audiology students.

University of Rochester 2020
Biomaterials Teaching Assistant · Rochester, New York
- Led lab sessions, review sessions, and office hours for biomaterials students.

Honors

- JSPS KAKENHI Grant-in-Aid for Research Activity Start-up (2026)
- FWO Strategic Basic Research Fellowship (2021-2025)
- Belgian American Educational Foundation Fellow (2021)
- Best "New Neuromorph" Award, Telluride Neuromorphic Cognition Engineering Workshop (2022)
- Fulbright Semifinalist for Open Study/Research Award to Belgium (2021)
- BME Graduate Student Teaching Assistant Award Honorable Mention (2020)
- Eagle Scout (2014)